



Unity Catalog

Unified and open governance for
data and AI

Speaker Name / Title



Governance is critical to scale your data and AI initiatives

“Organizations are finally realizing the value of **data as an asset** that needs to be protected, managed and maintained to **increase asset value**”



“Organizations seeing the **highest returns** from AI have a framework for **AI governance** to cover every step of the model development process”

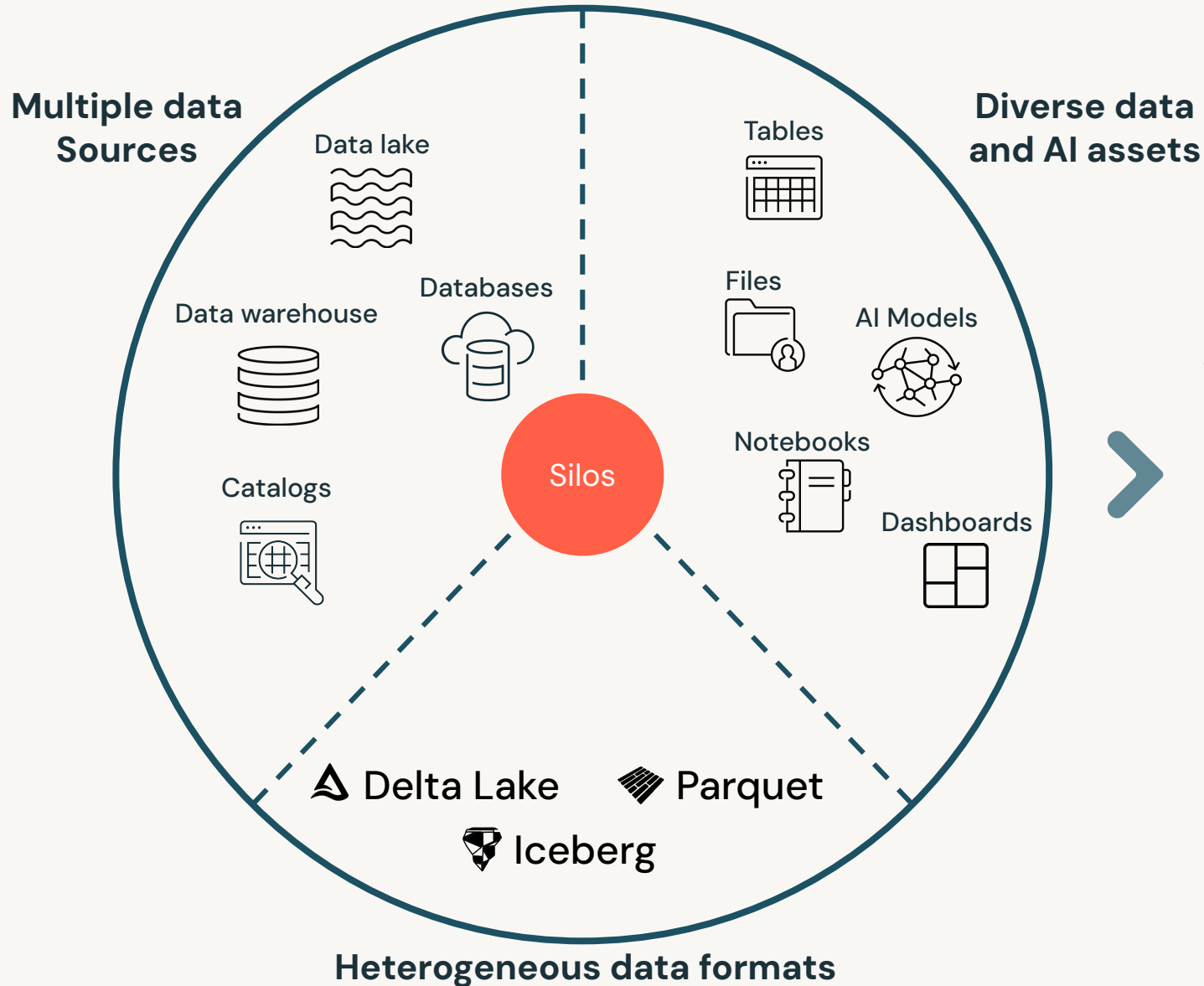
McKinsey
& Company

“AI is now an enterprise essential, and as such, **AI governance** will join cybersecurity and compliance as a **board-level topic**”

FORRESTER®



But governance of **the entire data estate** is hard



Fragmented governance for data and AI
Across access management, auditing, monitoring, observability, and lineage

Lack of open connectivity
For access and cross-platform collaboration

Lack of built-in intelligence
For data discovery, understanding, and gaining domain insights

These challenges **impact business value**

Fragmented governance



Compliance, security and quality risks
Operational inefficiencies

Lack of open connectivity



Vendor lock-in
Higher operational cost

Lack of built-in intelligence



Slow speed of innovation
Missed competitive edge



An **ideal governance** solution needs:

Unified Governance

All data and AI assets
and data formats

Open Connectivity

Any data source, compute
engine, and client

Built-in Intelligence

Powering domain knowledge
and platform optimizations



Unified and open governance for all data + AI



Traditional catalogs

Security

Access Control

Auditing

Collaboration

Discovery

Secure Data Sharing

Quality

Lineage

Quality Monitoring

Management

Cost Controls

Business Semantics

Tables



AI Models



Files



Notebooks



Dashboards



Delta Lake



Parquet



Iceberg

Connect to any external data source



Open access and collaboration with any tool, engine or platform

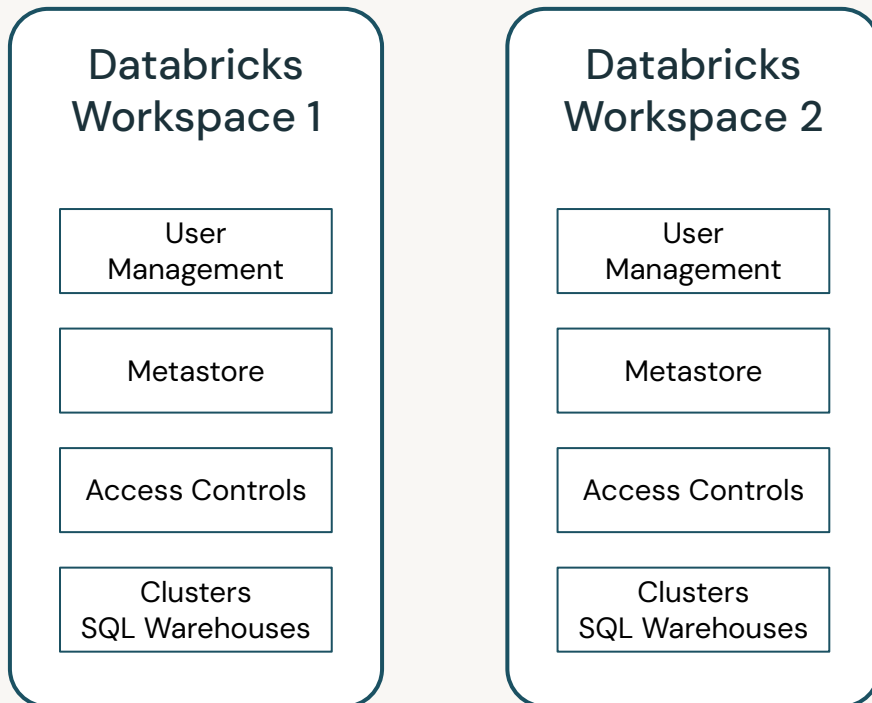


Unity Catalog Unlocks Full Potential of Your Lakehouse	w/ UC 	w/ Hive Metastore 
Governance across workspaces (Access controls, metastore, Identity management)	✓	✗ (Governance at workspace level)
Automated table and column level lineage in real-time across workspaces	✓	✗
Row Filters and Column Masking (use standard SQL functions for fine-grained access controls)	✓	✗
ABAC (Policies based on attributes and tags), RBAC (Role based policies)	✓	✗
Volumes (Govern non tabular data/arbitrary files – image, video, PDF etc.)	✓	✗
Lakehouse Federation (Discover, govern and query external databases, data warehouses, Hive Metastore, AWS Glue)	✓	✗
Lakehouse Monitoring (Monitor quality of data and ML models with auto generated alerts and dashboards)	✓	✗
Delta Sharing (Share files, tables, ML models, notebooks across clouds, regions and data platforms)	✓	✗
Business Semantics (Repository for managing and governing business metrics. Access from any tool any workload)	✓	✗
Databricks Marketplace	✓	✗
Databricks Cleanroom (Collaboration on data in a privacy-preserving manner)	✓	✗
AI/BI Genie spaces	✓	✗
Databricks Assistant (Generate and fix code in natural language)	✓	✗ (Degraded experience)
Governance for AI (manage and govern ML Models, Feature store)	✓	✗
System Tables (End to end observability for Billing, Auditing, Cost, Lineage, Marketplace analytics)	✓	✗
Databricks Apps	✓	✗
Databricks Connect V2	✓	✗
Lakebase	✓	✗
Predictive optimization (optimizes your query plans, data layout, intelligently balancing read vs. write performance)	✓	✗
Streaming Tables	✓	✗
Open APIs (Access UC data from other data platforms –Amazon Athena, Presto, Trino, EMR, Snowflake and more)	✓	✓
Vector Search for indexing	✓	✗
Serverless workflows	✓	✗

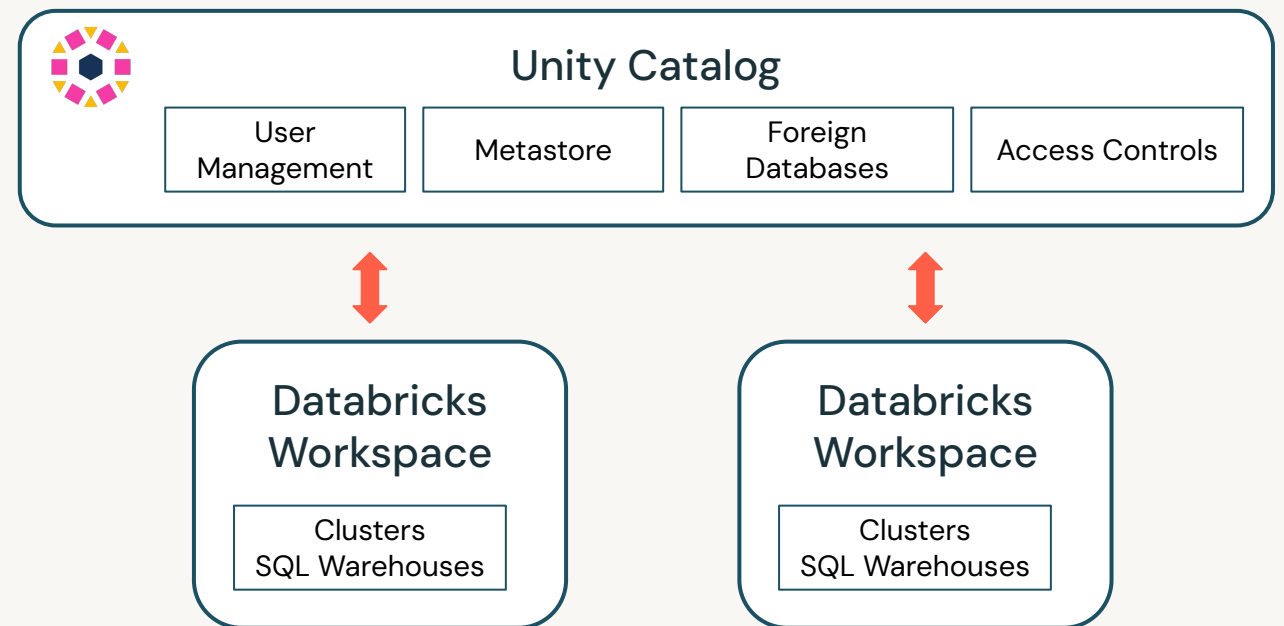
All your metadata, in one place

One metadata layer across file and database sources **superpowers** governance

Without Unity Catalog



With Unity Catalog



Fundamental Concepts

Working with file based data sources

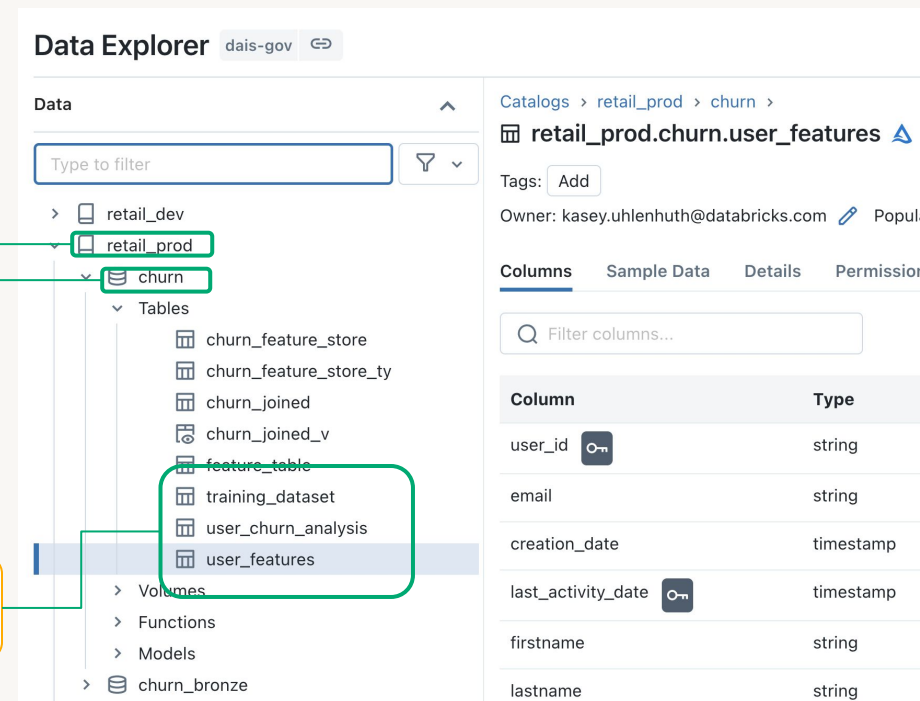
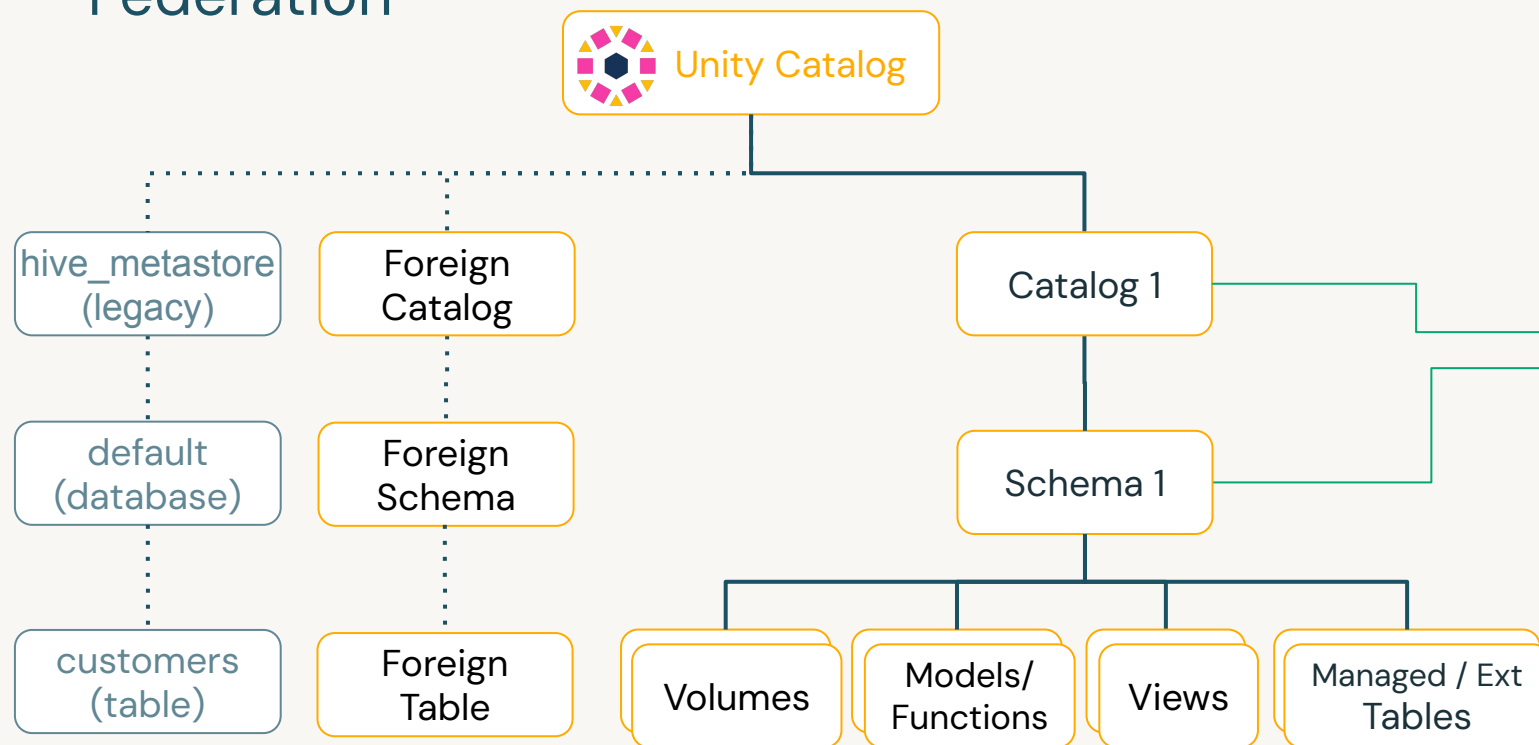
- **Credentials**
 - Storage Credential to connect to storage
 - Cloud Service Credential to connect native services
- **External Locations**
 - Storage location used for tables, files or volumes
- **Tables**
 - For managing tabular data
- **Volumes**
 - File container to govern a folder on cloud storage

Working with databases

- **Connections**
 - Credential and connection information to connect to an external database
- **Foreign Catalogs**
 - A catalog that represents an external database in UC and can be queried alongside managed data sources and file sources

Governed namespace across file and database sources

Access legacy metastore and foreign databases powered by Lakehouse Federation



```
SELECT * FROM main.paul.red_wine; -- <catalog>.<database>.<table>
```

```
SELECT * FROM hive_metastore.default.customers;
```

```
SELECT * FROM snowflake_warehouse.some_schema.some_table;
```

Centralized Access Controls

Centrally grant and manage access permissions across workloads

Using ANSI SQL DCL

```
GRANT <privilege> ON <securable_type>  
<securable_name> TO <principal>
```

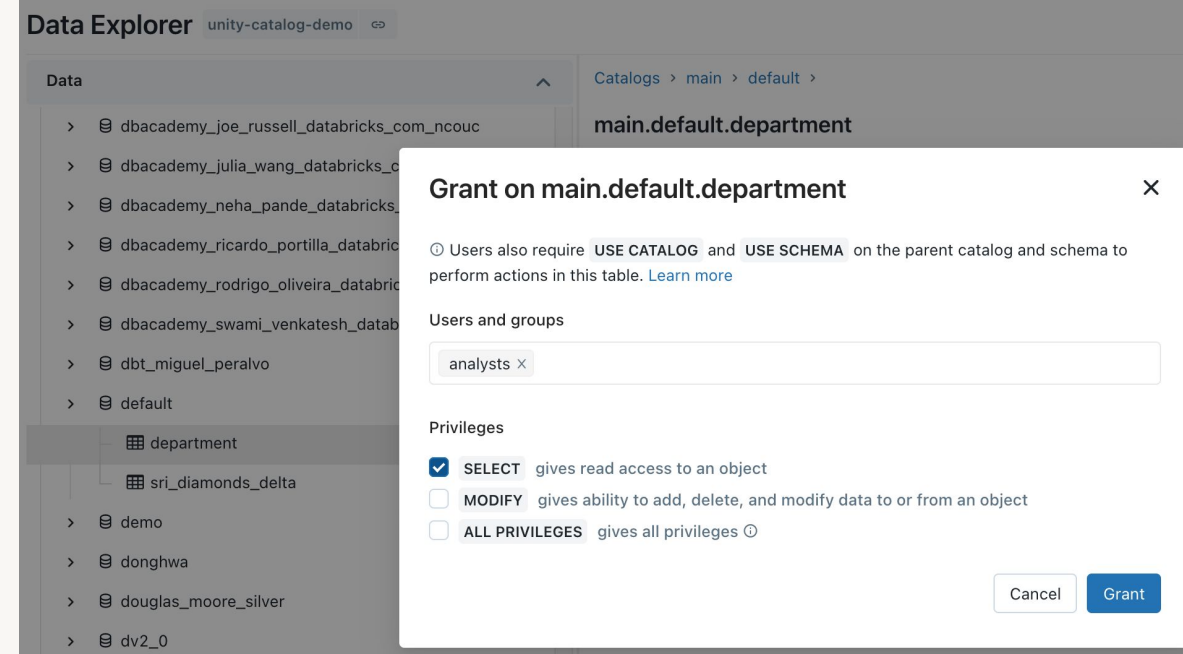
```
GRANT SELECT ON iot.events TO engineers
```

Choose
permission level

'Table' = collection of
files in S3/ADLS

Sync groups from
your identity
provider

Using UI



Row Level Security and Column Level Masking

Provide differential fine grained access to datasets

Only show specific rows

```
CREATE FUNCTION <name> ( <parameter_name >  
<parameter_type> .. )  
RETURN {filter clause whose output must be a boolean}
```

```
CREATE FUNCTION us_filter(region STRING)  
RETURN IF(is_account_group_member ('admin'), true,  
region="US");
```

```
ALTER TABLE sales SET ROW FILTER us_filter ON region;
```

Test for group at
the account level

Assign reusable
filter to table

Specify filter
predicates

Mask or redact sensitive columns

```
CREATE FUNCTION <name> (<parameter_name>,  
<parameter_type>, [, <column>...])  
RETURN {expression with the same type as the first  
parameter}
```

```
CREATE FUNCTION ssn_mask(ssn STRING)  
RETURN IF(is_account_group_member('admin'), ssn, "*****");
```

```
ALTER TABLE users ALTER COLUMN table_ssn SET MASK  
ssn_mask;
```

Test for group at
the account level

Assign reusable
mask to column

Specify mask or
function to mask

Working with Terraform & APIs

Use data-sec-ops, policies as code patterns to scale your efforts

- Privileges for UC objects can be managed programmatically using our Terraform provider, especially for teams already using Terraform
- This will pair naturally with the management of the UC objects (Metastore, Catalog, Assignments etc.) themselves.

(If not already using Terraform, maybe now is a good time!)

Documentation > Data governance guide > What is Unity Catalog? >
Automate Unity Catalog setup using Terraform

Automate Unity Catalog setup using Terraform

March 10, 2023

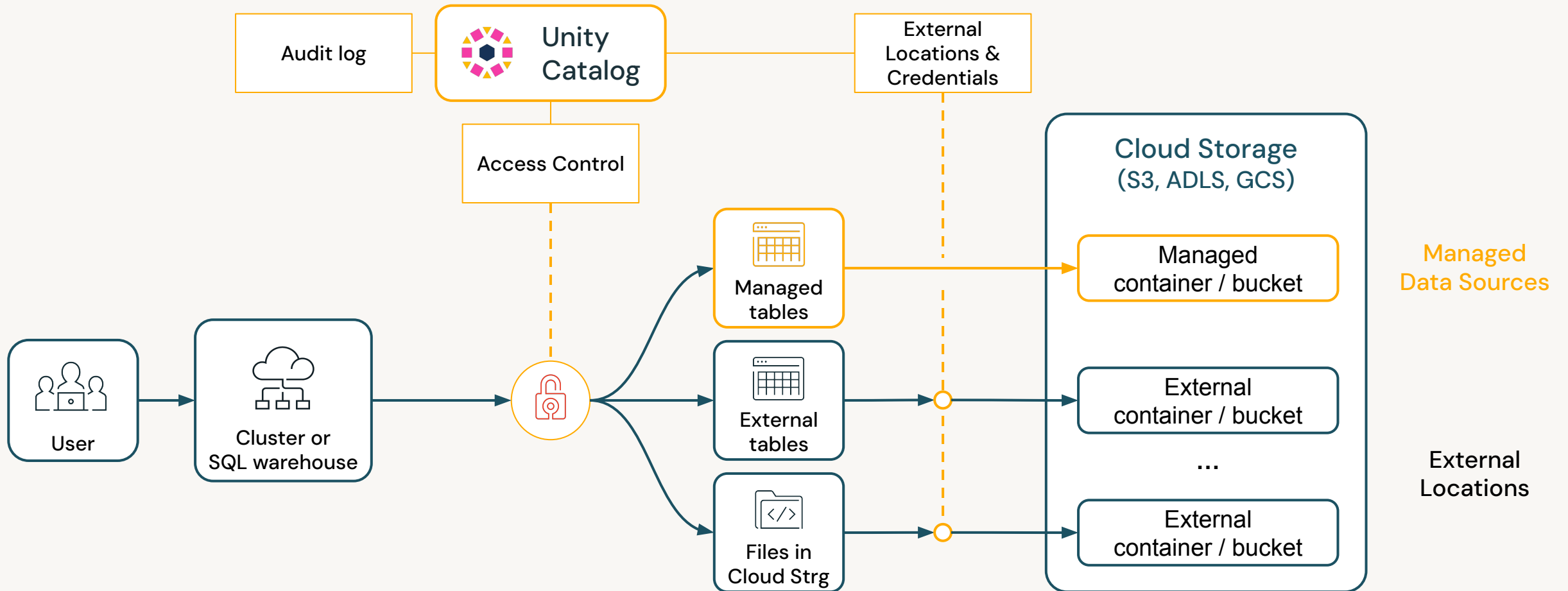
You can automate Unity Catalog setup by using the [Databricks Terraform provider](#). This article shows one approach to deploying an end-to-end Unity Catalog implementation. If you already have some Unity Catalog infrastructure components in place, you can also use this article to deploy additional Unity Catalog infrastructure components as needed.

For more information, see [Deploying pre-requisite resources and enabling Unity Catalog in the Databricks Terraform provider documentation](#).

```
resource "databricks_grants" "sandbox" {  
  provider = databricks.workspace  
  catalog = databricks_catalog.sandbox.name  
  grant {  
    principal = "Data Scientists"  
    privileges = ["USAGE", "CREATE"]  
  }  
  grant {  
    principal = "Data Engineers"  
    privileges = ["USAGE"]  
  }  
}
```

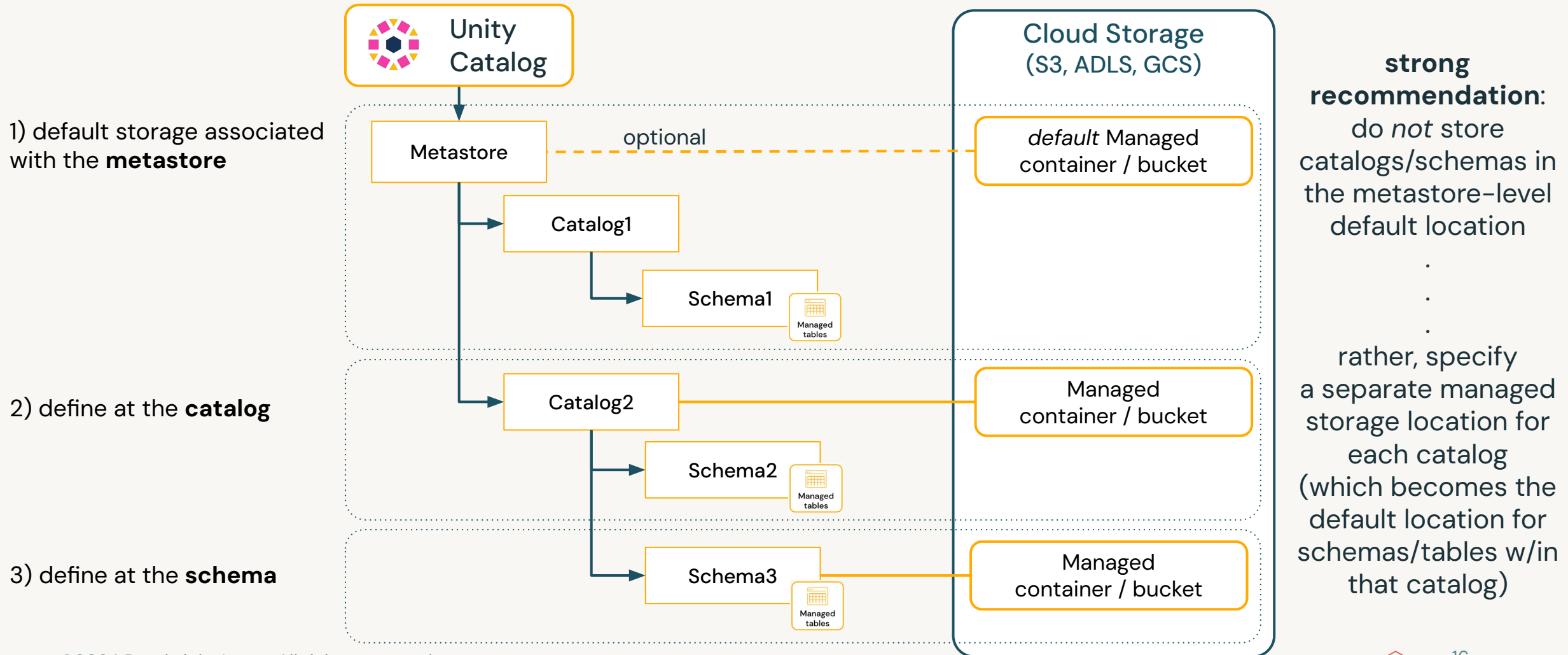
Managed Data Sources & External Locations

Simplify data access management across clouds



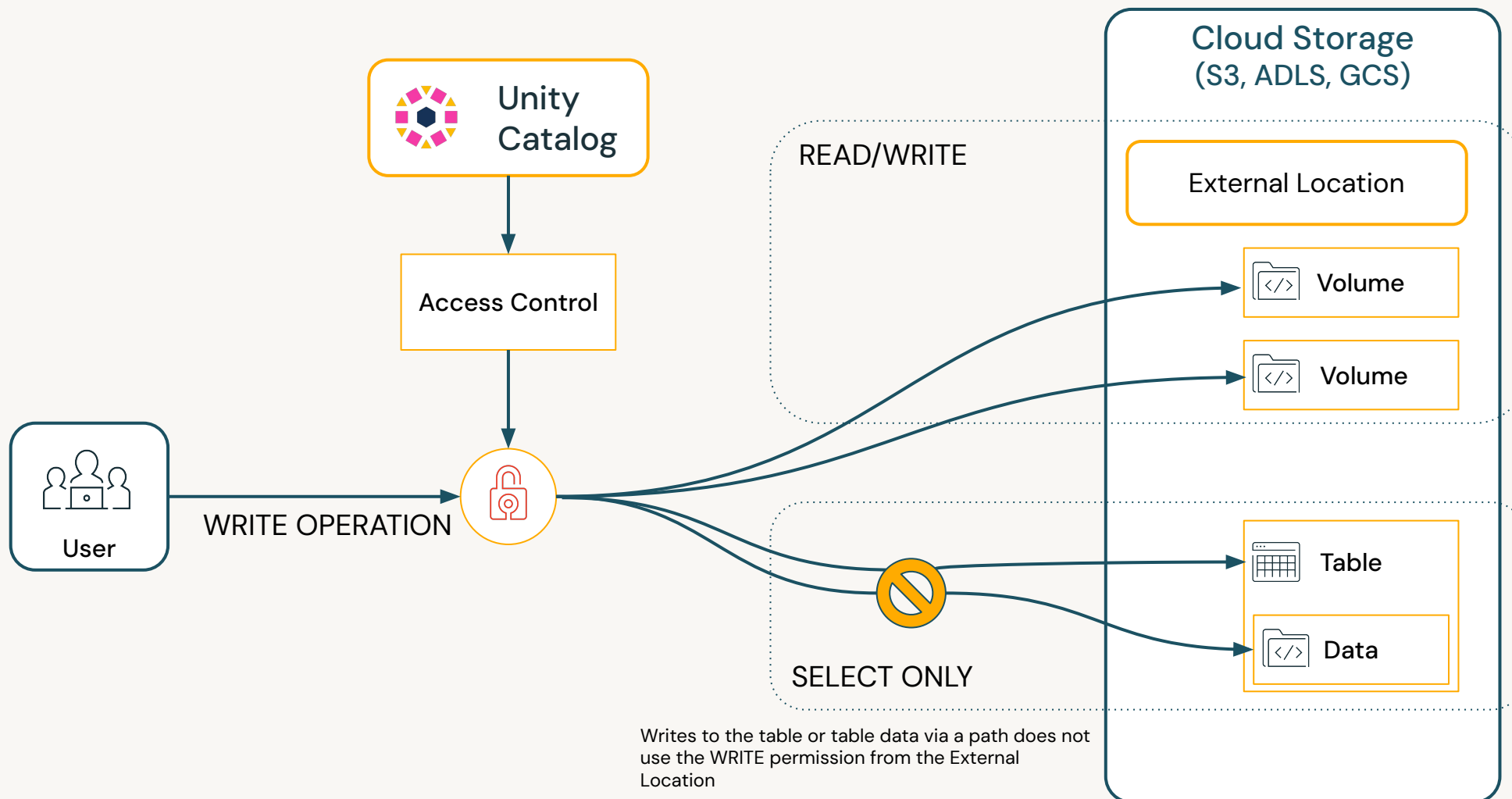
Default access to storage by catalog or schema

Use managed data sources for data isolation or cost allocation



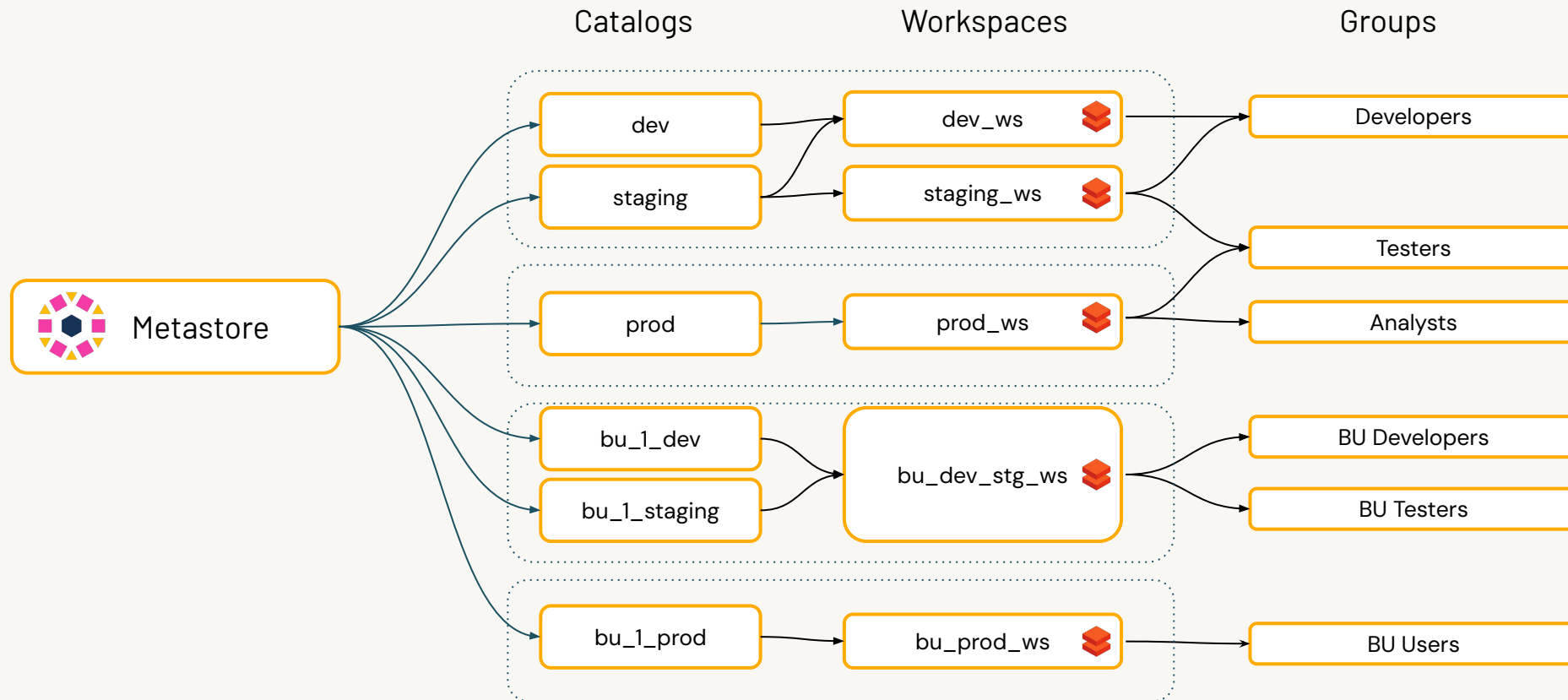
Govern filesystems and objects distinctly

Govern external tables and filesystem access separately



Access data from specified environments only

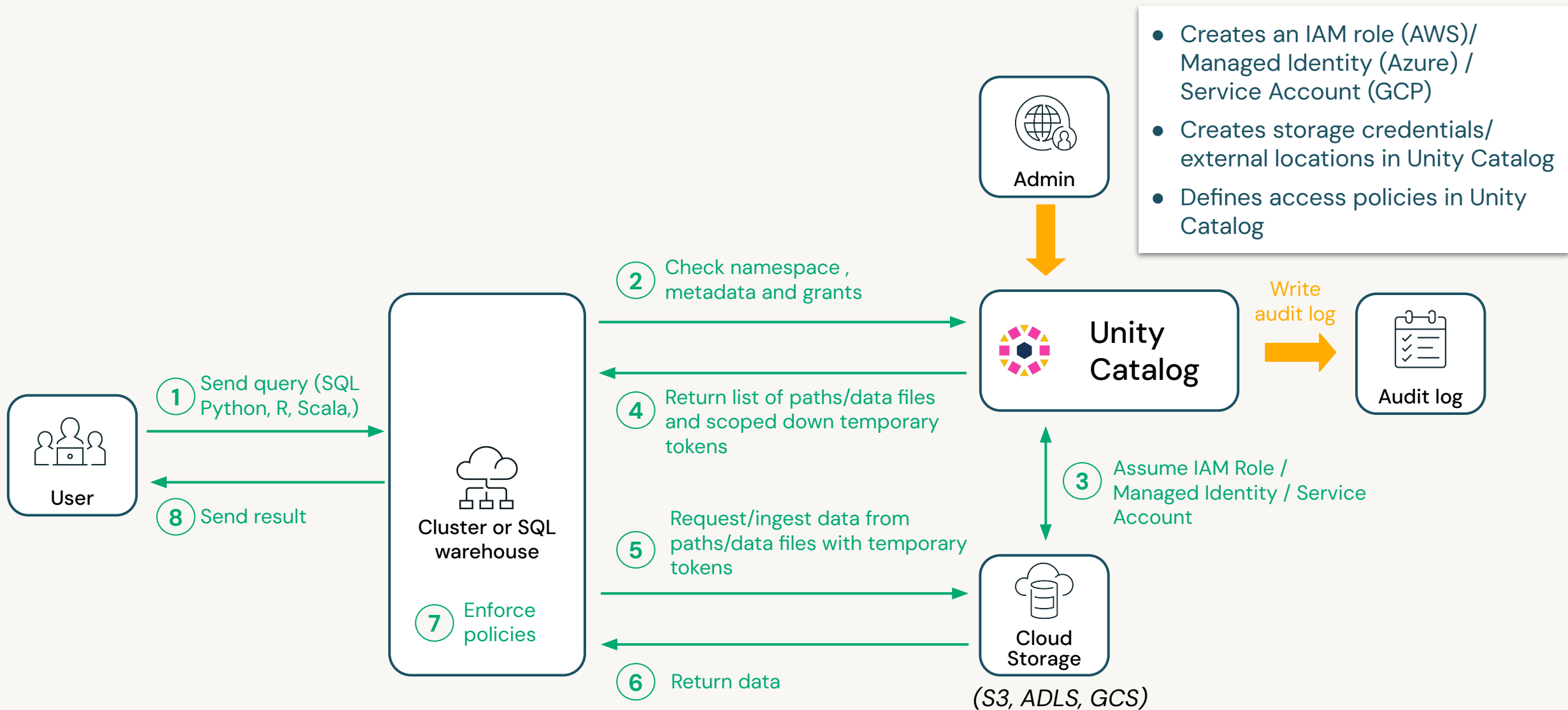
Restrict data access by environment or purpose



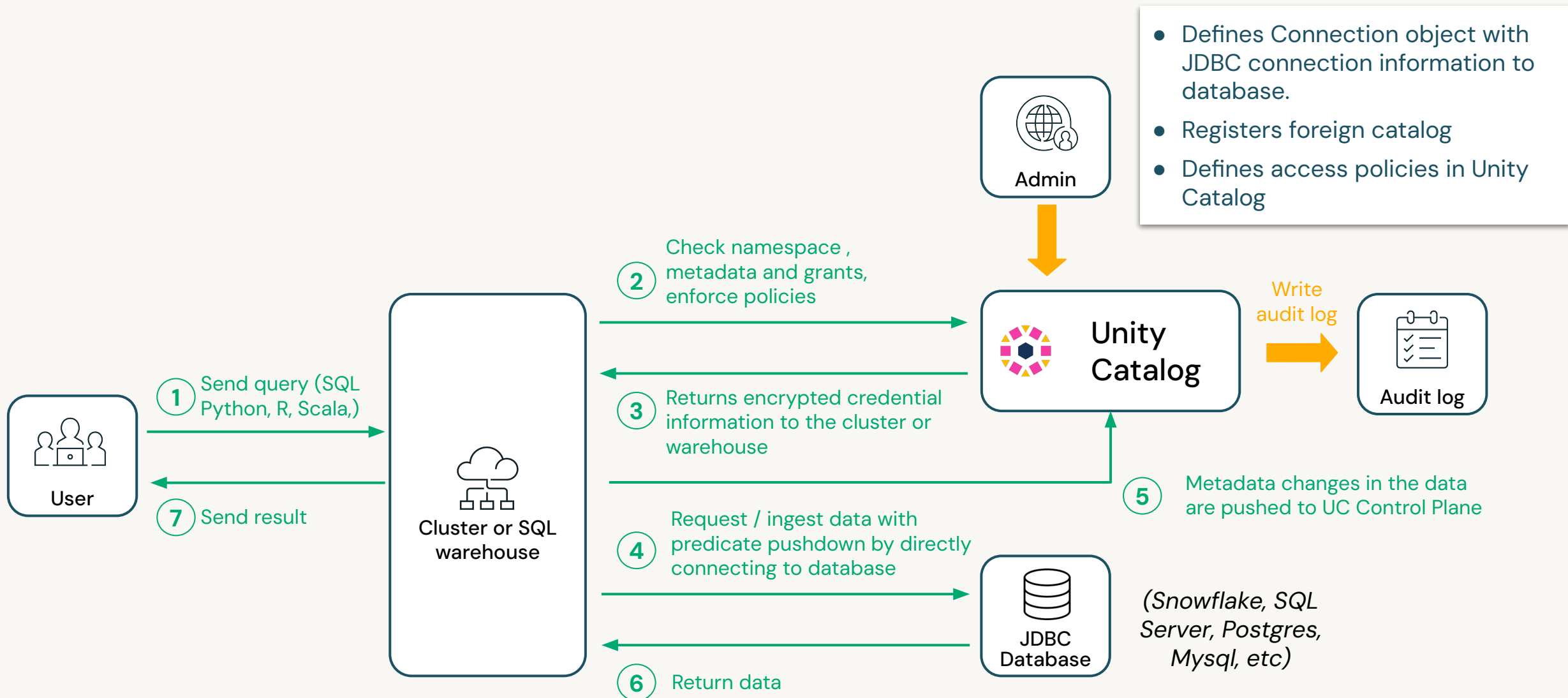
Access to data and availability of data can be isolated across workspaces and groups



Querying file based data sources with Unity



Querying database sources with Unity



Volumes in Unity Catalog

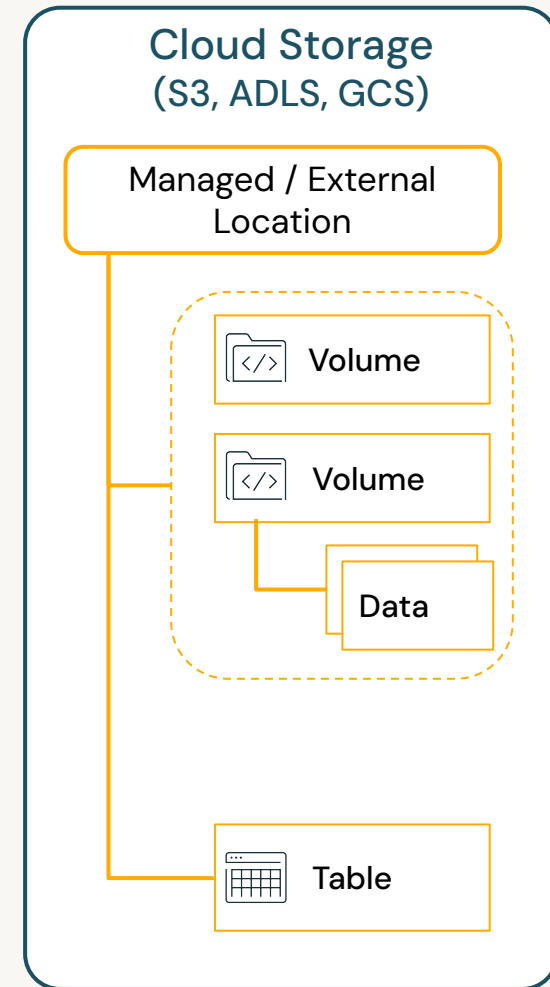
Access, store, organize and process files with Unity Catalog governance

- Volumes can be accessed by POSIX commands

```
dbutils.fs.ls("s3://my_external_location/Volumes/volume123")
```

```
ls /Volumes/volume123
```

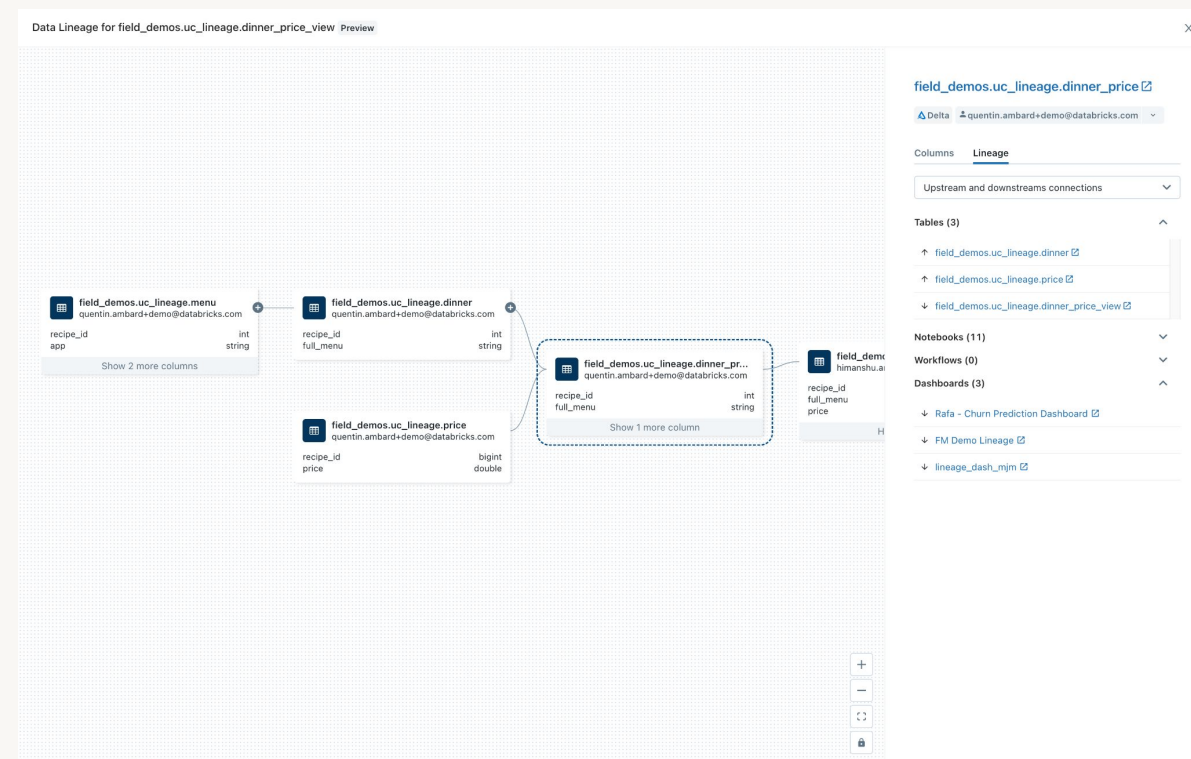
- Volumes are created under Managed or External Locations and show up in UC Lineage
- Volumes add governance over non-tabular data sets
 - Unstructured data, e.g., image, audio, video, or PDF files, used for ML
 - Semi-structured training, validation, test data sets, used in ML model training
 - Raw data files used for ad-hoc or early stage data exploration, or saved outputs
 - Library or config files used across workspaces
 - Operational data, e.g., logging or checkpointing output files
- Tables are registered in Managed / External Locations, not in Volumes



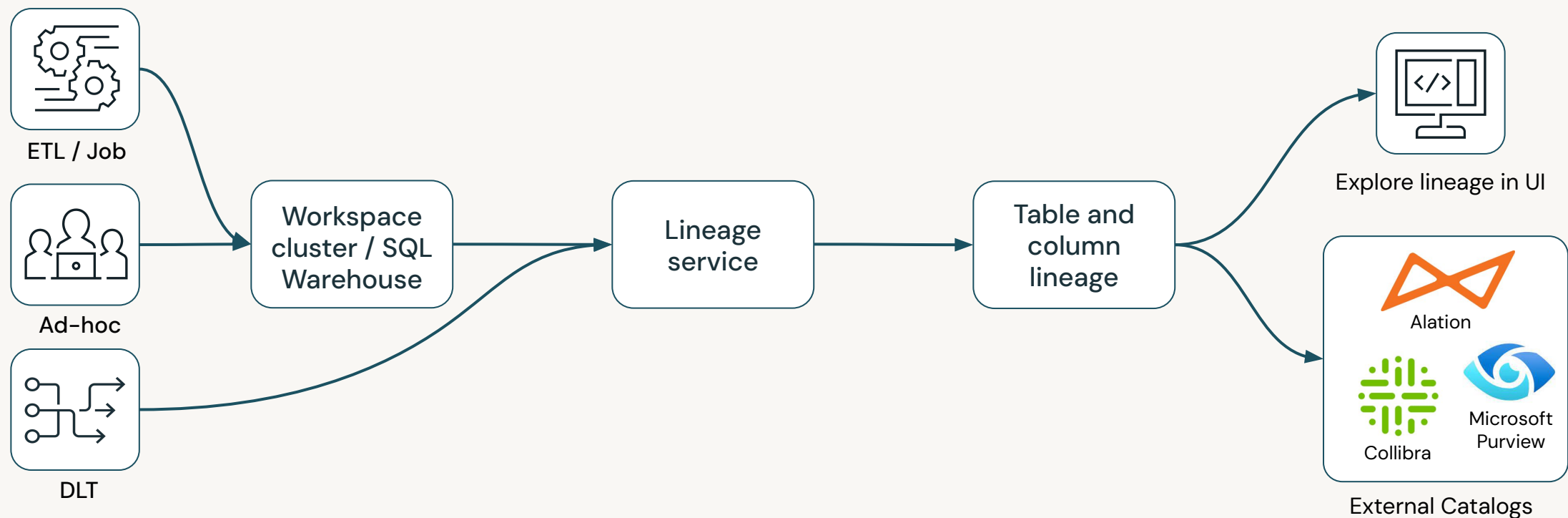
Automated lineage for all workloads

End-to-end visibility into how data flows and consumed in your organization

- Auto-capture runtime data lineage on a Databricks cluster or SQL warehouse
- Track lineage down to the table and column level
- Leverage common permission model from Unity Catalog
- Lineage across tables, dashboards, workflows, notebooks, feature tables, files, and DLT



Lineage flow – How it works



- Code (any language) is submitted to a cluster or SQL warehouse or DLT* executes data flow

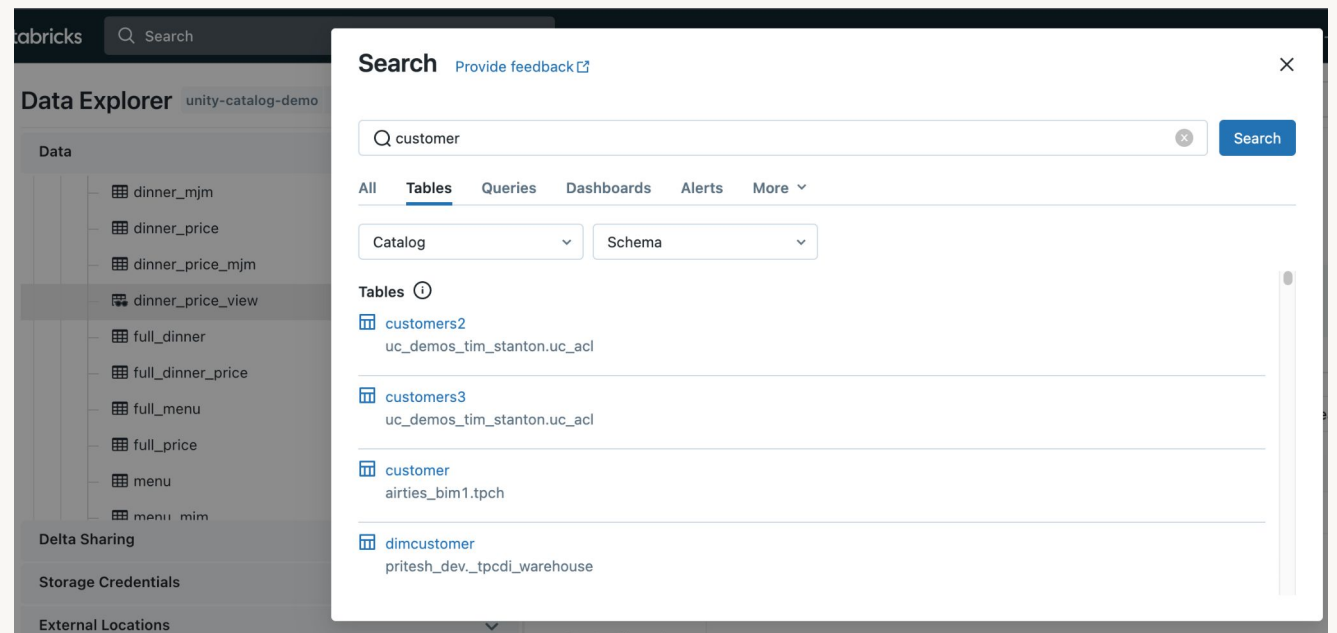
- Lineage service analyzes logs emitted from the cluster, and pulls metadata from DLT
- Assembles column and table level lineage

- Presented to the end user graphically in Databricks
- Lineage can be exported via API and imported into other tool

Built-in search and discovery

Accelerate time to value with low latency data discovery

- UI to search for data assets stored in Unity Catalog
- Unified UI across DSML + DBSQL
- Leverage common permission model from Unity Catalog
- Apply semantic tags to data and search across tags



Discovery Tags

Semantic layer for your lakehouse

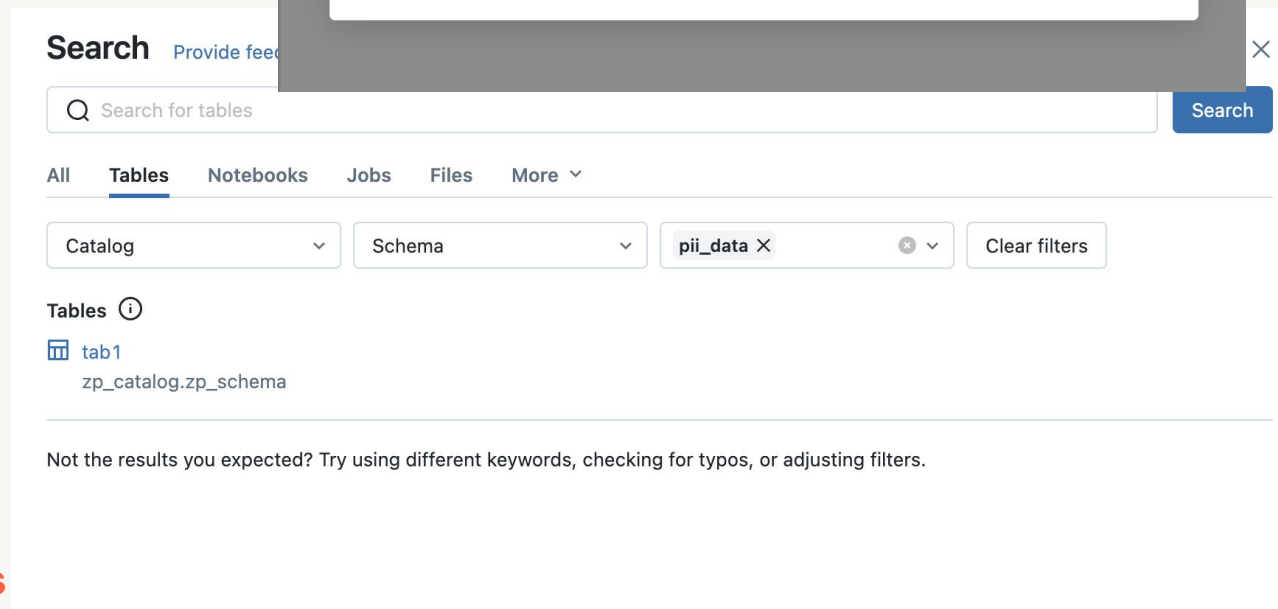
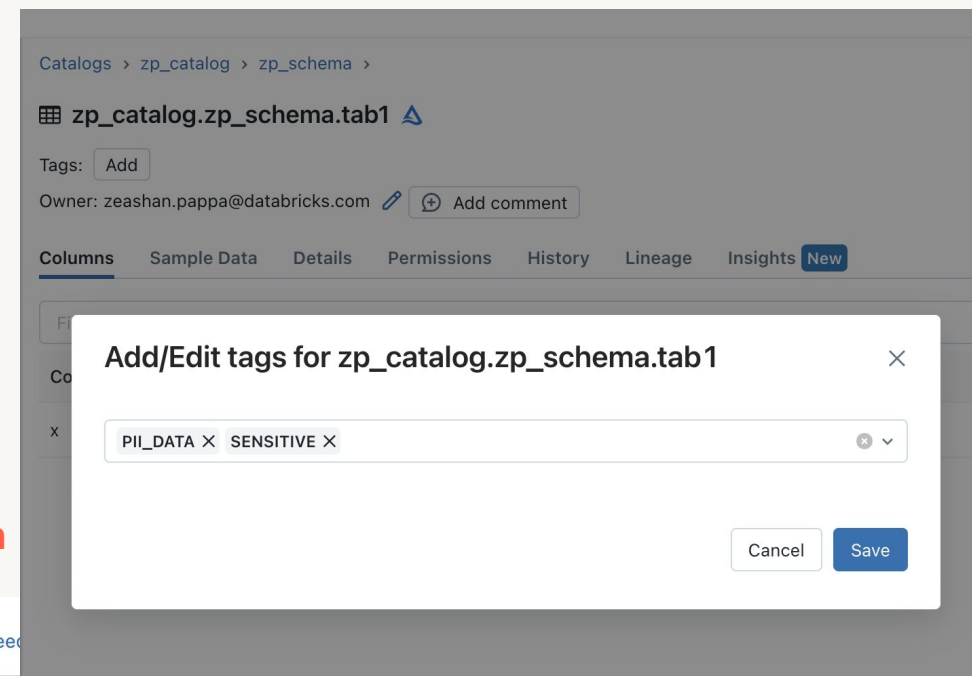
Problem

Searching for data assets in business terms or generally agreed upon taxonomies usually requires additional catalog tools.

Solution

- Discovery Tags allow you to tag Column, Table, Schema, Catalog objects in UC
- **Integrated search mechanism in UC allows you to search for objects by tag.**

Tag your data



Search based on tags